TECHNICAL REFERENCE GUIDE

Sharp AQUOS R7 Background Notification Fix

Complete Diagnostic Analysis & Permanent System Optimization Guide

Device: Sharp AQUOS R7 (A202SH)
OS Version: Android 14
Target Apps: WhatsApp Business, Snapchat & All Apps

1. Root Cause Breakdown

Through deep ADB inspection of Android system internals and Google Play Services, delayed notifications and the WhatsApp Business "single grey tick" were diagnosed as resulting from three distinct system failures:

A. WhatsApp Business: Missing FCM Push Token Registration
Inspecting `dumpsys activity service GcmService` revealed that while standard WhatsApp (`com.whatsapp:0`) and Snapchat (`com.snapchat.android:0`) were registered with Google FCM, `com.whatsapp.w4b` **was entirely missing**. The app was holding a stale token migrated from an older device and had never requested a fresh token from Google Play Services on the Sharp AQUOS R7. WhatsApp servers therefore never sent push packets to the phone during sleep.

B. Snapchat & General Apps: Android 14 Kernel Freezer & Netpolicy Firewall
Snapchat's background activity was suppressed by Android 14's kernel-level **Cached App Freezer** (`use_freezer=true`), which suspends background processes in Linux cgroups. Simultaneously, the system network policy whitelist omitted third-party app UIDs, cutting background socket access when the display was turned off.

2. System-Wide Fixes Applied via ADB

To eliminate background killing and network throttling across all present and future apps, the following permanent system changes were executed:

Step 1: Disable Android 14 Power & Standby Restrictors

```
# Disable App Standby Buckets completely
adb shell "settings put global app_standby_enabled 0"

# Disable Adaptive Battery throttling
adb shell "settings put global adaptive_battery_management_enabled 0"

# Keep Wi-Fi chip responsive during screen lock
adb shell "settings put global wifi_power_save 0"
adb shell "settings put global wifi_sleep_policy 2"

# Disable background data restriction
adb shell "cmd netpolicy set restrict-background false"
```

Step 2: Disable Linux Kernel App Freezer & Phantom Limiters

```
# Disable kernel cgroup process freezing
adb shell "device_config put activity_manager use_freezer false"

# Uncap phantom process killer
```

```
adb shell "device_config put activity_manager max_phantom_processes 2147483647"

# Lock device config against remote Google Phenotype overrides
adb shell "device_config set_sync_disabled_for_tests persistent"
```

Step 3: Whitelist All Installed 3rd-Party User Apps (Doze, Netpolicy, AppOps)

```
# Whitelist all user packages for Doze & AppOps execution
adb shell "for pkg in $(pm list packages -3 --user 0 | cut -d: -f2); do dumpsys deviceidle whitelist +$

# Add all user app UIDs to Network Policy Whitelist
adb shell "for uid in $(pm list packages -3 -U --user 0 | grep -o 'uid:[0-9]*' | cut -d: -f2 | sort -u)
```

3. Fixing WhatsApp Business Push Registration

> **Important Step:** Because WhatsApp Business has never registered an FCM token on this phone, perform this one-time clean setup:

1. **Back up your chats:** Open WhatsApp Business → **Settings** → **Chats** → **Chat backup** → tap **Back up**.
2. **Clear App Data or Reinstall:**
   - *Method A (Instant via ADB):* `adb shell pm clear com.whatsapp.w4b`
   - *Method B:* Uninstall WhatsApp Business and reinstall it from Google Play Store.
3. **Log in & Restore:** Open WhatsApp Business, verify your number, and restore your chat backup. The app will immediately acquire a valid FCM token from Google Play Services on this phone.

4. Persistence: Reboot vs. Factory Reset

Event	Impact on Configurations	Action Required
Normal Phone Reboot	**All fixes remain 100% active.** Global settings, network whitelist, and registered FCM push tokens survive standard restarts.	None. Everything continues working seamlessly.
Major OS Update (OTA)	Most settings remain, but kernel freezer or Phenotype configurations may revert to defaults.	Run `fix_all_notifications.bat` from PC.
Factory Reset	All customizations are wiped back to Japanese factory defaults.	Enable USB Debugging and run `fix_all_notifications.bat`.

5. One-Click Automation Script (`fix_all_notifications.bat`)

An automated script has been created in your platform-tools folder:

```
Location: c:\Users\Abdullah_\Desktop\platform-tools\fix_all_notifications.bat
```

Whenever you reset your phone or install a batch of new communication apps, simply plug your device in and double-click this file to reapply all system-level exemptions in 5 seconds.

6. Diagnostic Verification Commands

Use these commands to verify notification status at any time:

```
# 1. Open Google FCM Diagnostics screen
adb shell am start -n com.google.android.gms/.gcm.GcmDiagnostics

# 2. Check registered apps in Google FCM queue
adb shell "dumpsys activity service GcmService | grep -A 25 'Apps supporting client queue:'"

# 3. Check if an app is exempted from battery optimization
adb shell "dumpsys deviceidle whitelist | grep com.whatsapp.w4b"

# 4. Check if app UID has unrestricted network policy access
adb shell cmd netpolicy list restrict-background-whitelist
```